AP PRECALCULUS

Unit 1 • Lesson 1.6 practice • Answer key

Lesson 1.6 • Polynomial Functions and End Behavior

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to \infty} \left(-4x^3 + x^2\right)$ is

(A) ∞

(B) $-\infty$

(C) -4

(D) 0

Solution

Answer: (B). The leading term $-4x^3$ has odd degree and negative coefficient.

Question 2

NO CALCULATOR

Let $p(x) = (2 - x)^3$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$

(B) $+\infty$

(C) 8

(D) 0

Solution

Answer: (B). The leading term is $(-x)^3 = -x^3$. As $x \to -\infty$, $-x^3 \to +\infty$.

Question 3

NO CALCULATOR

Which polynomial increases without bound as $x \to \infty$ and as $x \to -\infty$?

(A) $x^3 + 1$

(B) $-x^4$

(C) $5x - x^2$

(D) $2x^6 - x$

Solution

Answer: (D). Even degree with a positive leading coefficient.

Question 4

NO CALCULATOR

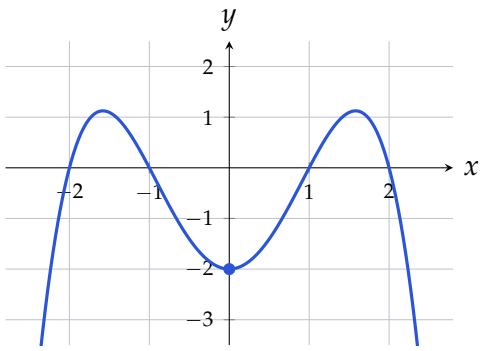
The graph of a polynomial falls to the left and rises to the right. Its degree and leading coefficient are

(A) even, positive
(B) even, negative
(C) odd, positive
(D) odd, negative

Solution
Answer: (C). Opposite end behaviors mean odd degree; rising to the right means positive leading coefficient.

Question 5 (graph) NO CALCULATOR

The graph of a polynomial r is shown. Its zeros are ± 1 and ± 2, and $r(0) = -2$. Which could be $r(x)$?



(A) $x^4 - 5x^2 + 4$
(B) $-x^3 + 4x$
(C) $-\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$
(D) $\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$

Solution
Answer: (C). Both ends of the graph go down: even degree, negative leading coefficient. Only (C) has these zeros, a negative leading coefficient, and $r(0) = -\frac{1}{2}(4) = -2$.

Question 6 (Free response) NO CALCULATOR

Let $f(x) = -2x^4 + 3x^3 + x - 7$ and $g(x) = x^2(3 - x)(x + 1)^2$.

(a) Write limit statements for the end behavior of f.
(b) Find the degree and leading term of g, and write limit statements for its end behavior.
(c) Could the graph of g have 5 turning points? Explain.

Solution
(a) Even degree, negative leading coefficient: $\lim_{x\to\infty} f(x) = -\infty$ and $\lim_{x\to-\infty} f(x) = -\infty$.
(b) Leading term $x^2 \cdot (-x) \cdot x^2 = -x^5$, degree 5. $\lim_{x\to\infty} g(x) = -\infty$ and $\lim_{x\to-\infty} g(x) = +\infty$.
(c) No. A polynomial of degree 5 has at most 4 turning points.